

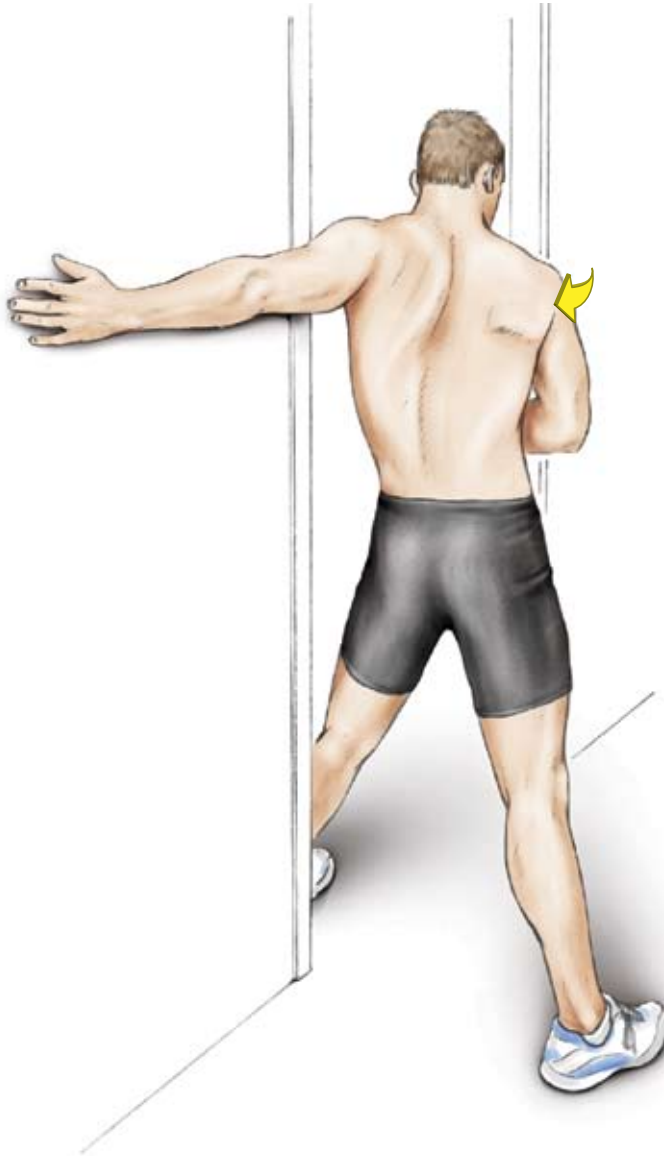
## ARMS, WRISTS, AND HANDS

The major joint of the arm, the elbow, is a hinge and thus has only the capacity to either flex or extend. As a result, the muscles that flex the elbow (biceps brachii, brachialis, brachioradialis, pronator teres) are located anteriorly, whereas the extensor muscles (anconeus, triceps brachii) are located posteriorly. The forearm contains two bones: radius and ulna. The radius gets its name from its ability to roll over the ulna, and this ability allows the palm to face either forward (supinated) or backward (pronated). There are two muscles that supinate (biceps brachii and supinator) and two muscles that pronate (pronator teres and pronator quadratus). The pronator muscles are located so that they can pull the radius toward the center of the body, and the supinator muscles are situated to pull the radius away from the body. Interestingly, most of the muscles that control wrist, hand, and finger movements are located at or near the elbow. This results in the belly of the muscle lying near the elbow with tendons crossing the wrist and attaching to the wrist (carpal), hand (metacarpal), and finger bones. Having only tendons in the wrists and hands prevents the wrists and hands from getting too bulky from the increase in size that accompanies muscle strength. Similar to the muscles that move the elbow, all of the wrist flexors (flexor carpi radialis, flexor carpi ulnaris, and palmaris longus) and most of the finger flexors (flexor digitorum profundus, flexor digitorum superficialis, and flexor pollicis longus) are located in the anterior compartment of the forearm. In contrast, all of the wrist extensors (extensor carpi radialis brevis, extensor carpi radialis longus, extensor carpi ulnaris, extensor digitorum communis) and finger extensors (extensor digitorum communis, extensor digiti minimi, extensor indicis) are located in the posterior compartment of the forearm. The muscles that run along the radius (which have *radialis* in their name) do ulnar deviation, whereas those along the ulna (which have *ulnaris* in their name) do radial deviation. Just before crossing the wrist, the tendons of these muscles are anchored firmly by thick tissue bands called the flexor retinacula and extensor retinacula. By passing under the retinacula at the carpal (wrist bones), the tendons lie in a carpal tunnel. Since the tendons are crowded together, each tendon is surrounded by a slippery sheath to minimize friction. Illustrations showing these muscles as well as a chart showing specific movements for each muscle are located at the end of the chapter (pages 50-52).

Stretching the muscles that move the elbows and wrists is helpful in alleviating and sometimes preventing overuse injuries. Because it is more resistive to opposing movements, a tight muscle is easy to damage. When the wrist flexor muscles are tight, pain arises on the medial (close to the body) side of the elbow. In sports, this pain is sometimes referred to as “golf elbow.” Tight wrist extensor muscles, on the other hand, can cause pain on the opposite, or lateral, side of the elbow. This pain is frequently called “tennis elbow.” Also, constant wrist hyperextension can lead to overstretched tendons in the carpal tunnel. This causes the wrist flexor muscles to tighten, and the constant contraction can lead to increased friction, inflammation, and overuse injury (carpal tunnel syndrome). Continuous stretching of the wrist flexors can strengthen the tendons and help alleviate future problems.

Many of the instructions and illustrations in this chapter are given for either the left or right side of the body. Similar but opposite procedures would be used for the opposite (not pictured) side of the body.

## Elbow Flexor Stretch



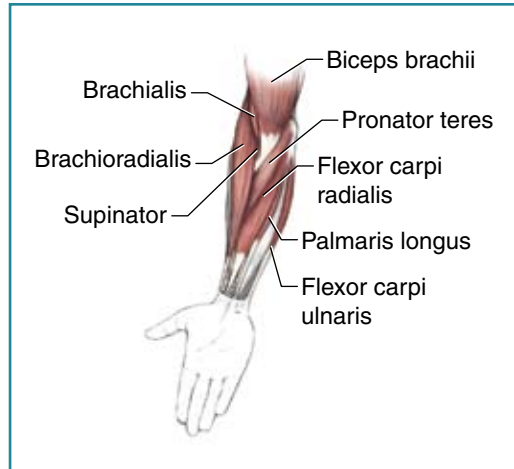
### Technique

Stand in a doorway.

While keeping the arm straight, raise the left arm to shoulder level.

Place the arm and palm against the wall with the thumb pointing up.

Rotate the trunk backward toward the wall.



## Muscles Stretched

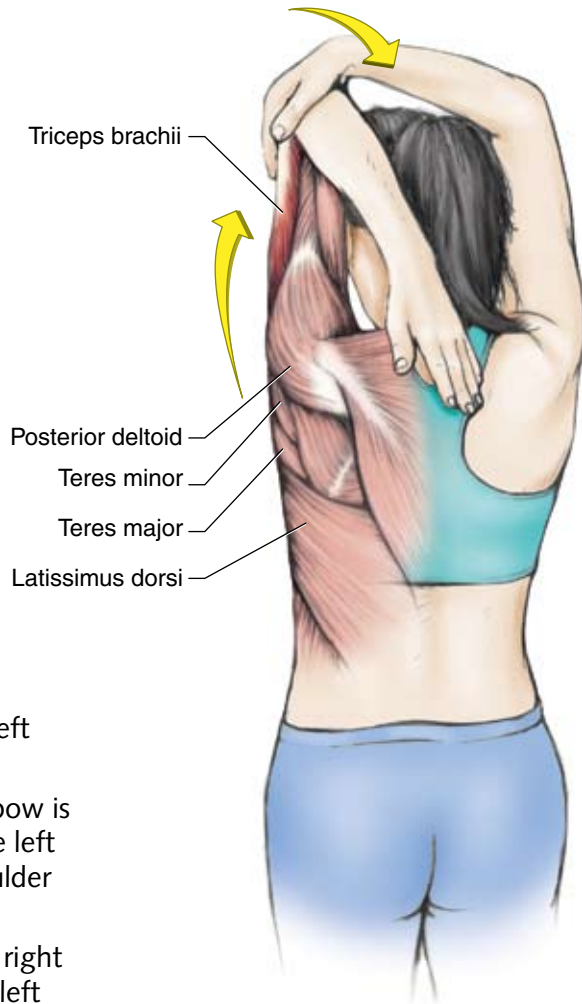
**Most-stretched muscles:** Left brachialis, left brachioradialis, left biceps brachii.

**Lesser-stretched muscles:** Left supinator, left pronator teres, left flexor carpi radialis, left flexor carpi ulnaris, left palmaris longus.

## Commentary

This stretch is easier to do by grasping a solidly fixed vertical pole. A tight grasp, however, virtually eliminates the stretch effect on the lesser-stretched muscles. Also, it is more difficult to keep the elbow straight, and a straight elbow is necessary for this stretch to be effective. Although it is preferable to lift the arm to shoulder level, the stretch will be effective at whatever height the arm is raised.

# Elbow Extensor (Triceps Brachii) Stretch



## Technique

- Sit or stand upright with the left arm flexed at the elbow.
- Raise the left arm until the elbow is next to the left ear and the left hand is near the right shoulder blade.
- Grasp the left elbow with the right hand and pull or push the left elbow behind the head and toward the floor.

## Muscles Stretched

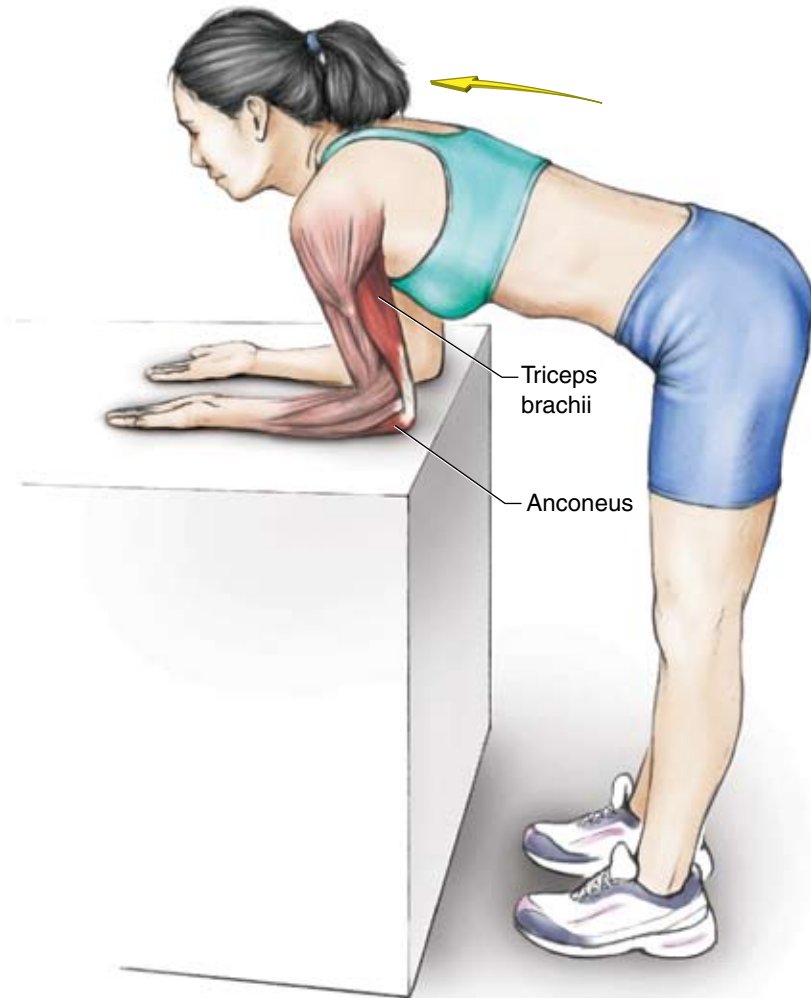
**Most-stretched muscle:** Left triceps brachii.

**Lesser-stretched muscles:** Left latissimus dorsi, left teres major, left teres minor, left posterior deltoid.

## Commentary

Doing this stretch while seated in a chair with a back allows better control of balance. A greater stretching force can be applied to the muscles when the body is balanced.

## Elbow Extensor (Anconeus) Stretch



### Technique

Stand or sit upright while facing a table.

Flex the elbows and rest the forearms on the table with the palms up.

Lean forward, bringing the chest toward the table.

### Muscles Stretched

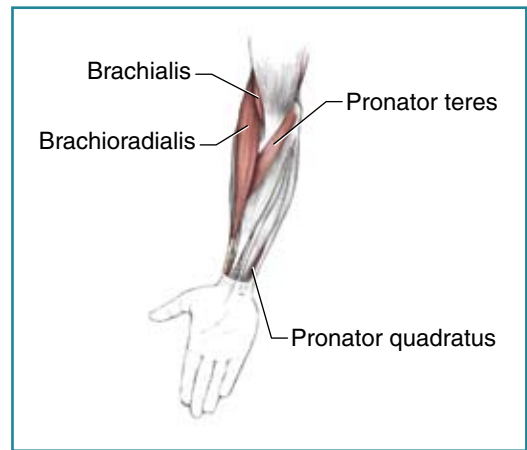
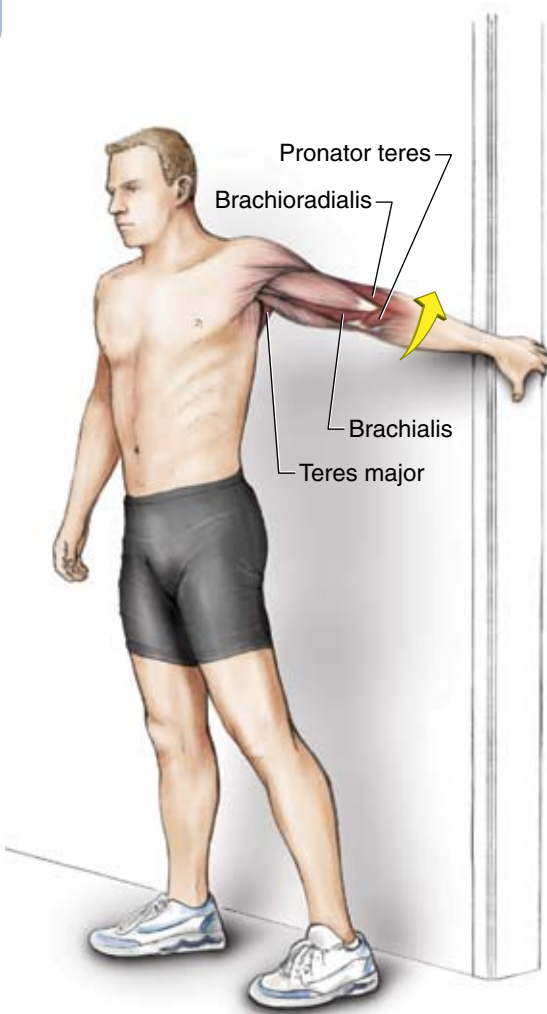
**Most-stretched muscle:** Left anconeus.

**Lesser-stretched muscle:** Left triceps brachii.

### Commentary

For the greatest stretch, keep the forearms and elbows flat on the table.

## Forearm Pronator Stretch



### Technique

Stand with the back toward the inside of the doorframe.

While keeping the arm straight, hyperextend the left arm above the midpoint between the hip and shoulder.

Grasp the doorframe with the left hand with the thumb pointing down.

Externally rotate the arm (roll the biceps toward the top).

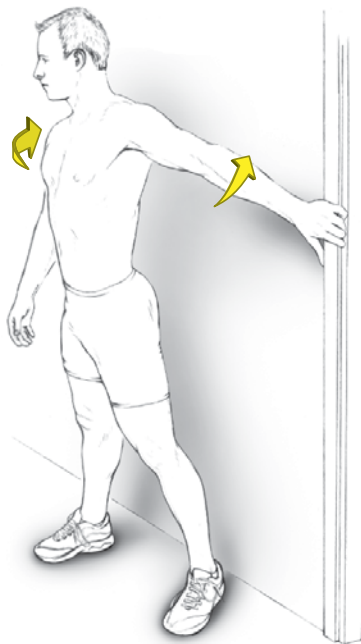
## Muscles Stretched

**Most-stretched muscle:** Left pronator teres.

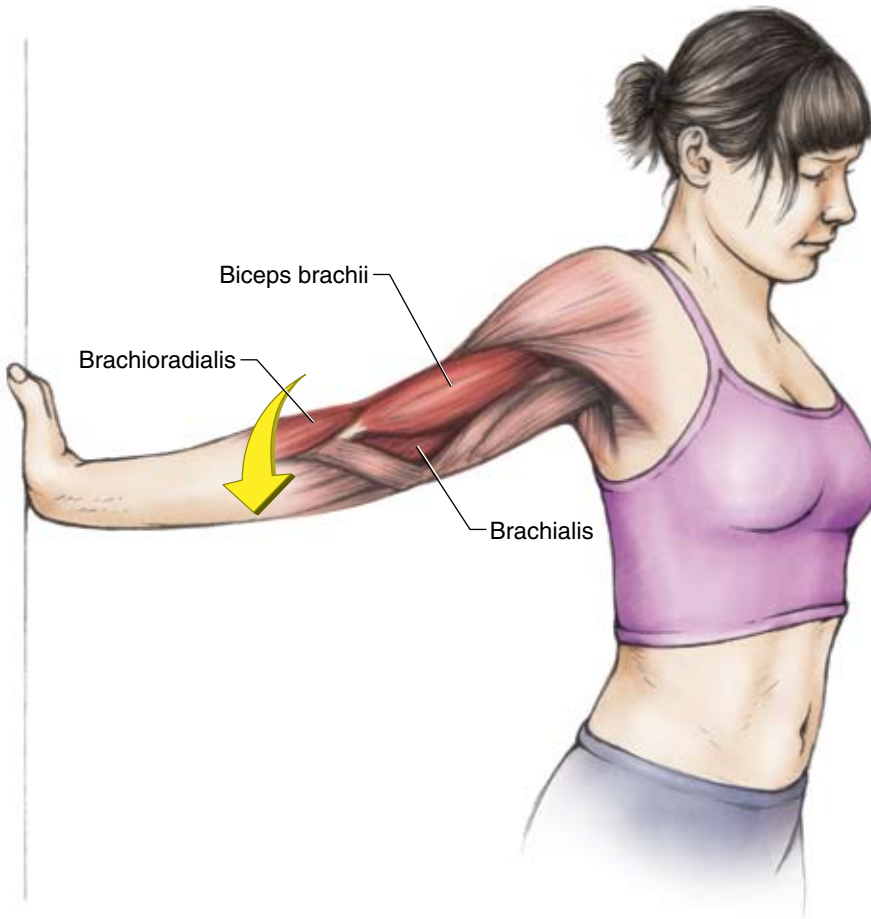
**Lesser-stretched muscles:** Left brachialis, left brachioradialis, left pronator quadratus, left subscapularis, left teres major.

## Commentary

You can also do this exercise with a firmly planted vertical pole. To maximize the stretch, keep the elbow straight. After rolling the biceps upward, you can enhance the stretch by inwardly rotating the back toward the hyperextended arm (see figure).



## Forearm Supinator Stretch



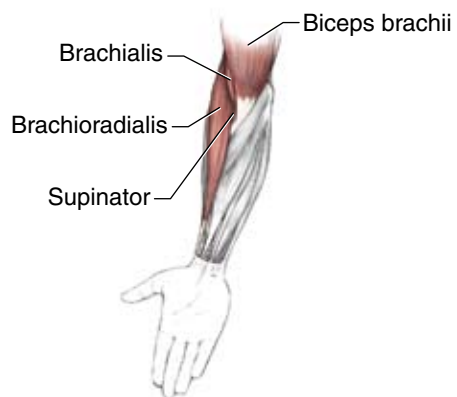
### Technique

Stand with the back toward the inside of the door-frame.

While keeping the arm straight, hyperextend the right arm above the mid-point between the hip and shoulder.

Grasp the doorframe with the right hand with the thumb pointing up.

Internally rotate the arm (roll the biceps down).



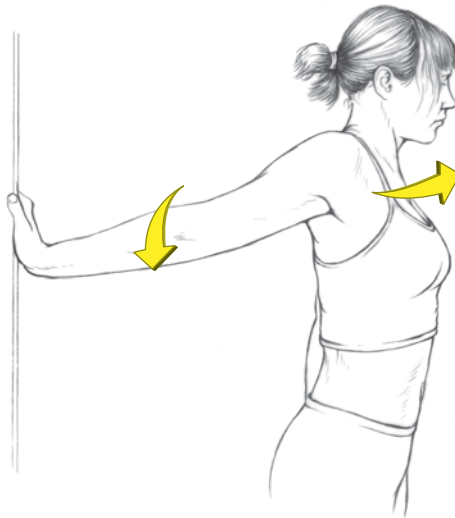
## Muscles Stretched

**Most-stretched muscles:** Right biceps brachii, right supinator.

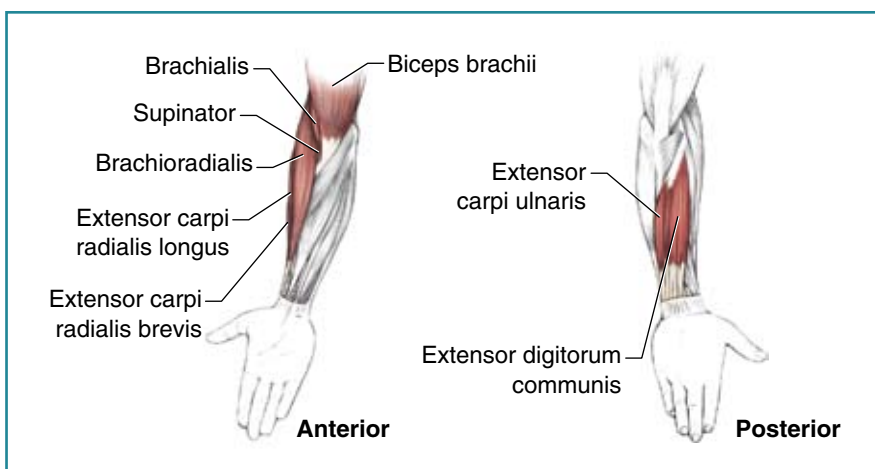
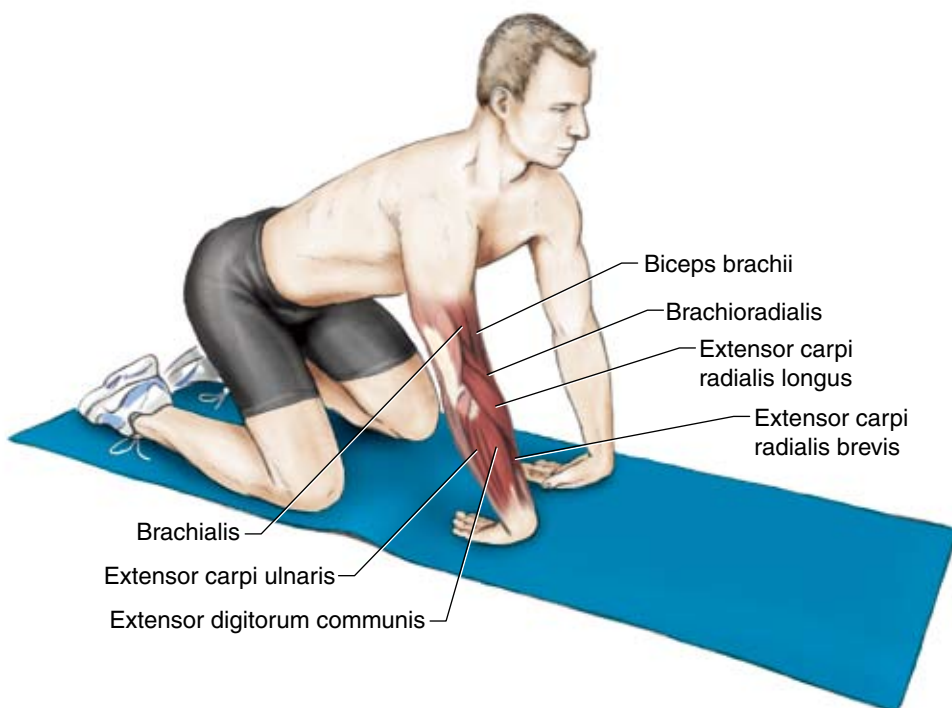
**Lesser-stretched muscles:** Right brachialis, right brachioradialis, right infraspinatus, right teres minor.

## Commentary

You can also do this exercise with a firmly planted vertical pole. To maximize the stretch, keep the elbow straight. After rolling the biceps downward, you can enhance the stretch by inwardly rotating the back toward the hyperextended arm (see figure).



## Wrist Extensor Stretch



## Technique

Kneel on the floor.

Flex both wrists and place the back of each hand on the floor, shoulder-width apart.

Point the fingers toward the knees.

While keeping the elbows straight, lean backward (buttocks to the heels), keeping the backs of the hands on the floor.

## Muscles Stretched

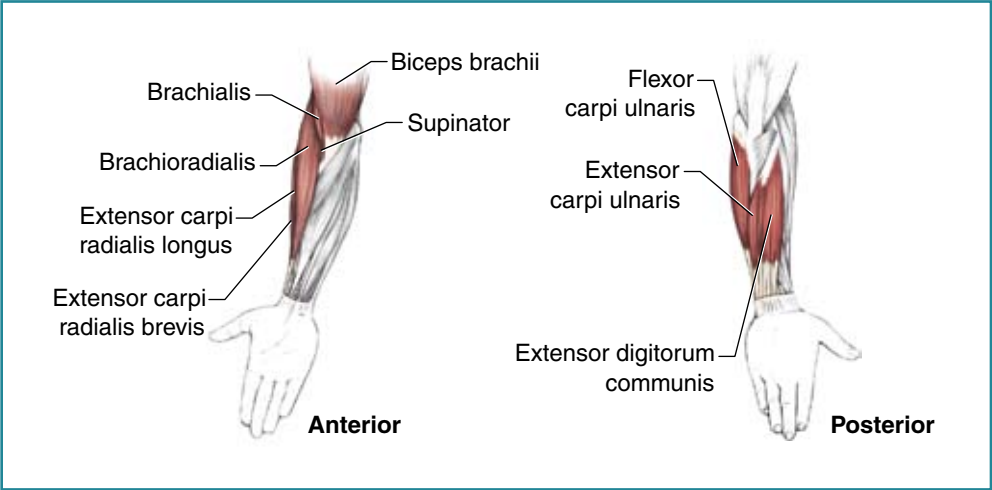
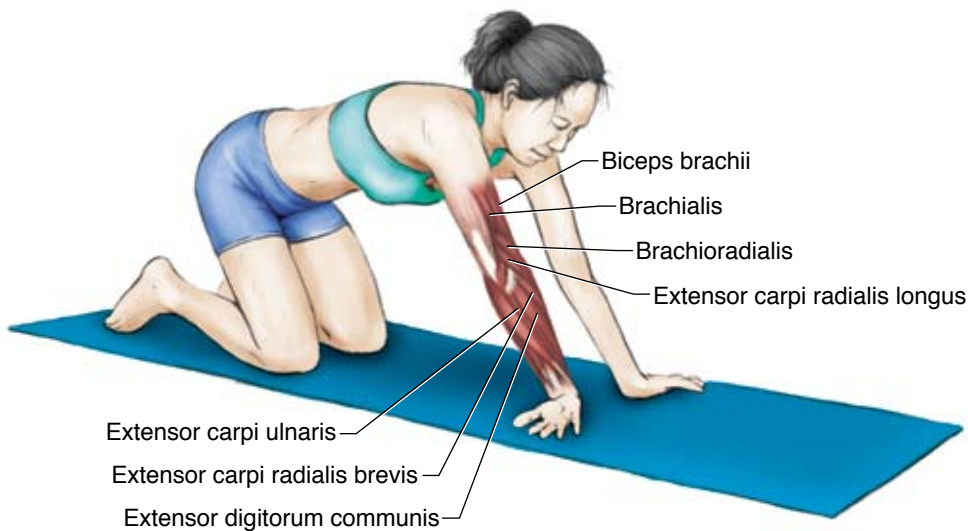
**Most-stretched muscles:** Brachioradialis, extensor carpi radialis brevis, extensor carpi radialis longus, extensor carpi ulnaris.

**Lesser-stretched muscles:** Supinator, brachialis, biceps brachii, extensor digitorum communis.

## Commentary

The closer the hands are to the knees, the easier it is to keep the backs of the hands touching the floor. The farther the hands are in front of the knees, however, the greater the applied stretch.

# Wrist Ulnar Deviator and Extensor Stretch



## Technique

Kneel on the floor.

Flex both wrists and place the back of each hand on the floor.

Point the fingers laterally on a line perpendicular to the midline of the body (the fingertips of the opposing hands pointing away from each other).

While keeping the elbows straight, lean backward (buttocks to the heels), keeping the backs of the hands on the floor.

## Muscles Stretched

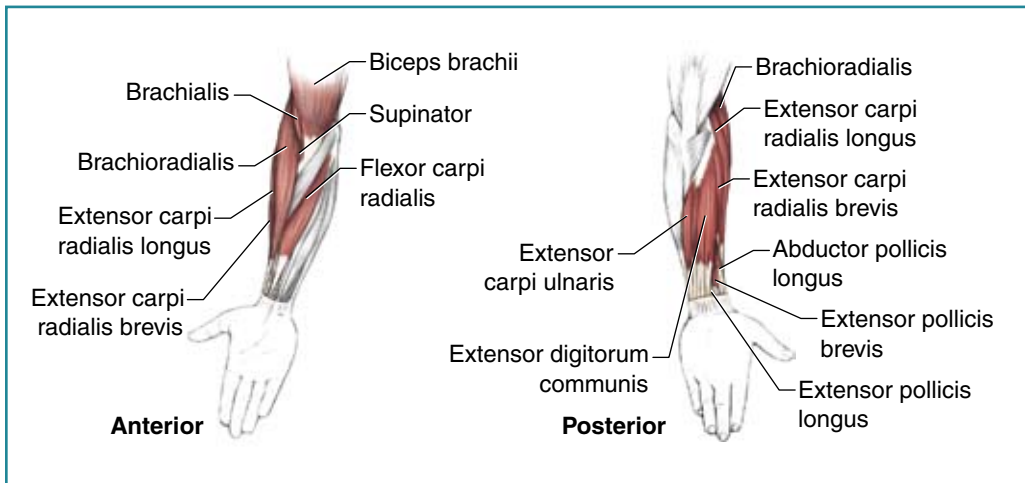
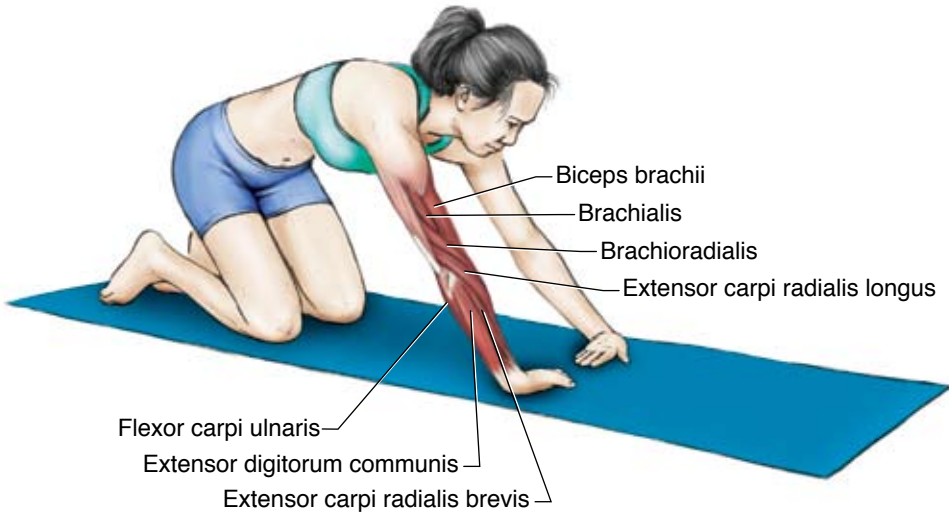
**Most-stretched muscles:** Extensor digitorum communis, extensor pollicis brevis, extensor carpi ulnaris.

**Lesser-stretched muscles:** Extensor carpi radialis brevis, extensor carpi radialis longus, extensor pollicis longus, flexor carpi ulnaris, brachioradialis, supinator, brachialis, biceps brachii.

## Commentary

The closer the hands are to the knees, the easier it is to keep the backs of the hands touching the floor. The farther the hands are in front of the knees, however, the greater the applied stretch. The distance each hand is away from the body's midline also influences stretch intensity. The farther away from the midline, the greater the stretch.

## Wrist Radial Deviator and Extensor Stretch



## Technique

Kneel on the floor.

Flex both wrists and place the back of each hand on the floor.

Point the fingers medially (the fingertips of the opposing hands pointing toward each other).

While keeping the elbows straight, lean backward (buttocks to the heels), keeping the backs of the hands on the floor.

## Muscles Stretched

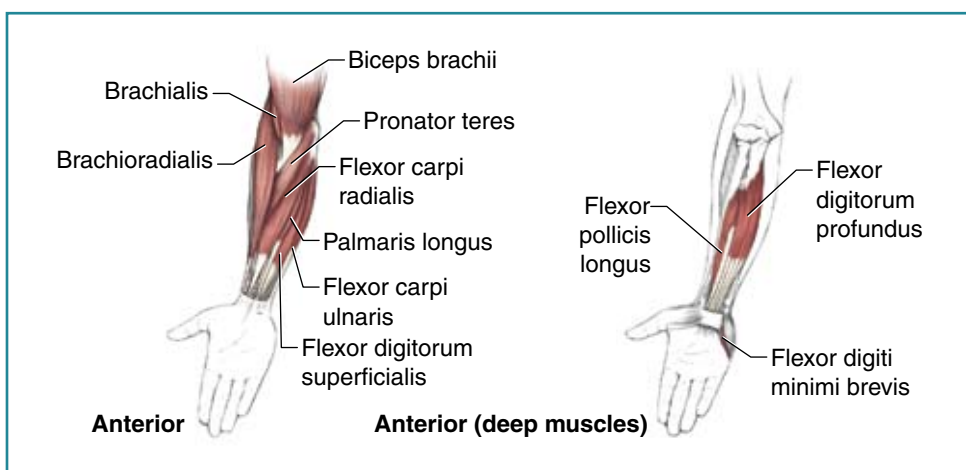
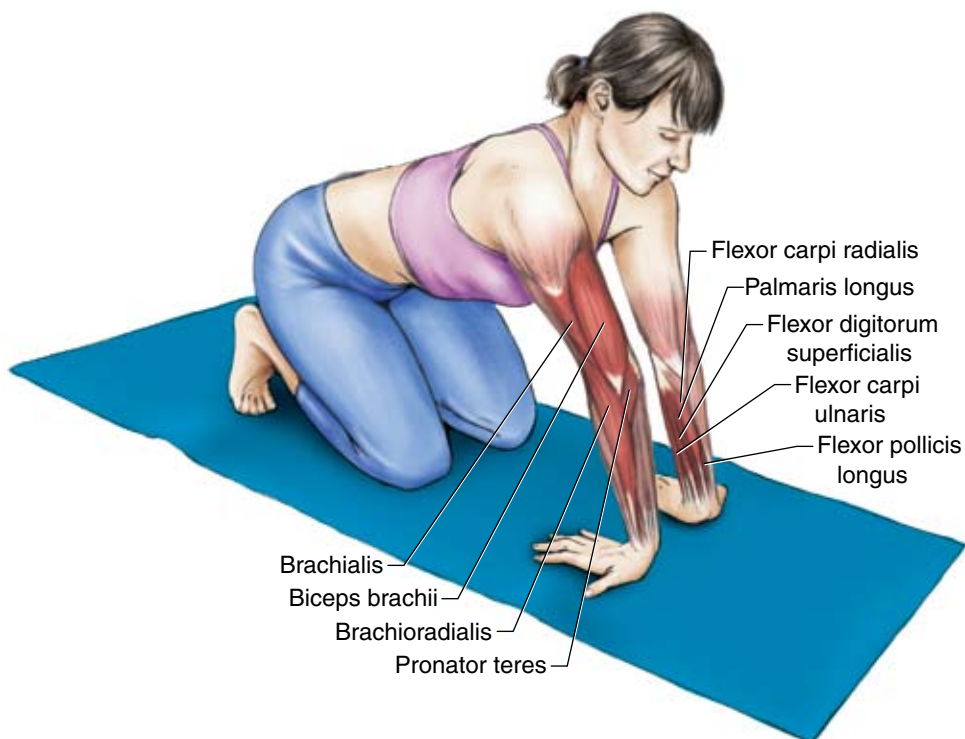
**Most-stretched muscles:** Extensor carpi radialis brevis, extensor carpi radialis longus, extensor digitorum communis, extensor pollicis brevis.

**Lesser-stretched muscles:** Extensor carpi ulnaris, flexor carpi radialis, supinator, brachialis, biceps brachii, brachioradialis.

## Commentary

The closer the hands are to the knees, the easier it is to keep the backs of the hands touching the floor. The farther the hands are in front of the knees, however, the greater the applied stretch. The distance each hand is away from the body's midline also influences stretch intensity. The farther away from the midline, the greater the stretch.

## Wrist Flexor Stretch



## Technique

Kneel on the floor.

Flex both wrists and place the palm of each hand on the floor, shoulder-width apart.

Point the fingers toward the knees.

While keeping the elbows straight, lean backward (buttocks to the heels), keeping the palms flat on the floor.

## Muscles Stretched

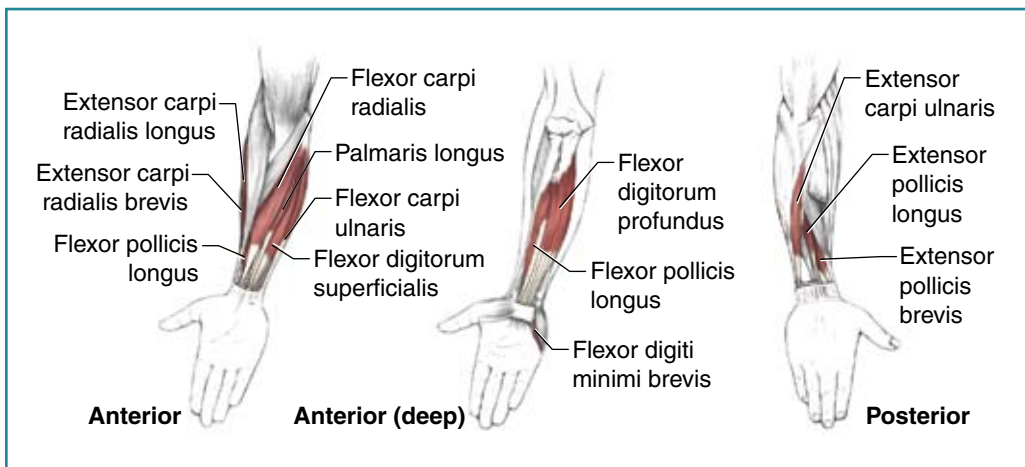
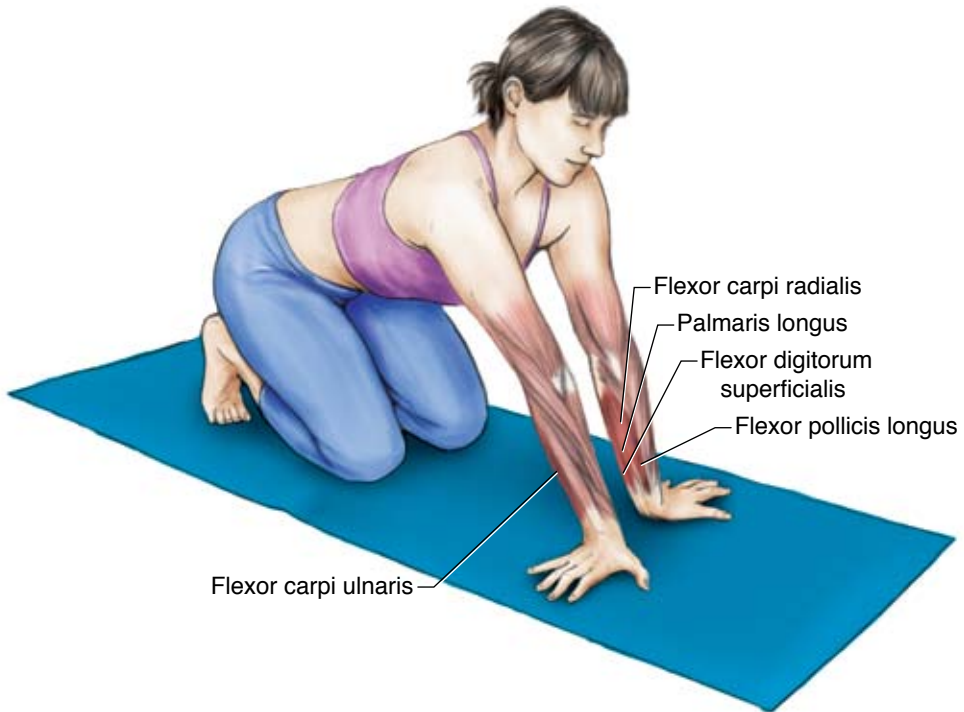
**Most-stretched muscles:** Brachioradialis, flexor carpi radialis, flexor carpi ulnaris, flexor digitorum profundus, flexor digitorum superficialis, palmaris longus.

**Lesser-stretched muscles:** Flexor digiti minimi brevis, flexor pollicis longus, pronator teres, brachialis, biceps brachii.

## Commentary

The closer the hands are to the knees, the easier it is to keep the backs of the hands touching the floor. The farther the hands are in front of the knees, however, the greater the applied stretch.

## Wrist Radial Deviator and Flexor Stretch



## Technique

Kneel on the floor.

Flex both wrists, and place the palm of each hand on the floor.

Point the fingers outward on a line perpendicular to the midline of the body.

While keeping the elbows straight, lean backward (buttocks to the heels), keeping the palms flat on the floor.

## Muscles Stretched

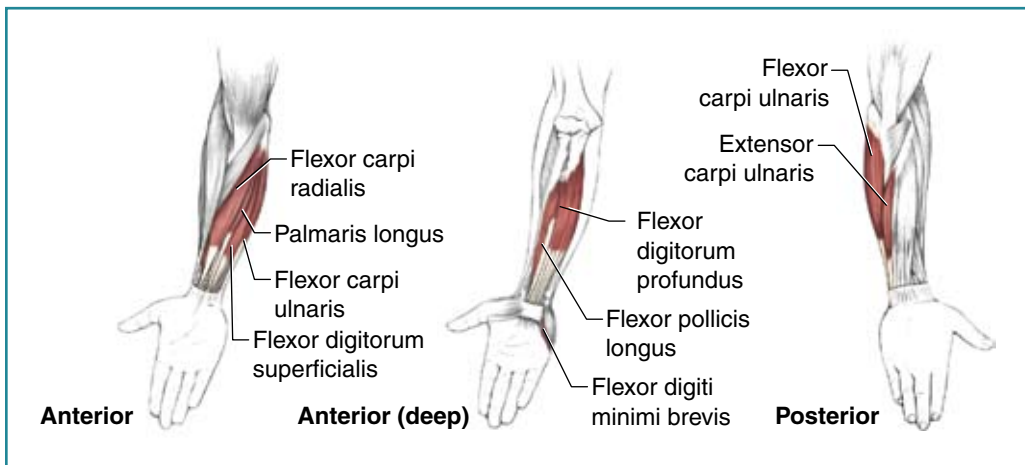
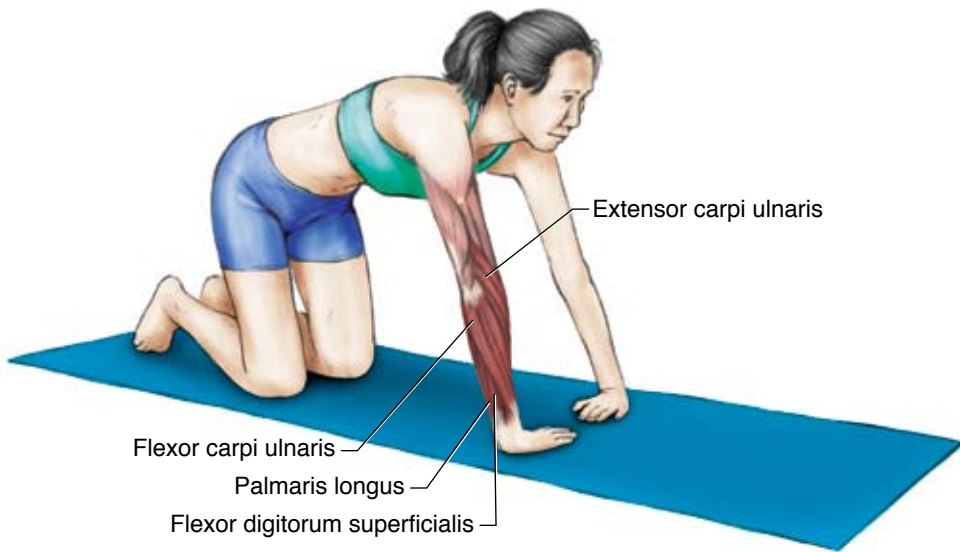
**Most-stretched muscles:** Flexor carpi radialis, flexor digitorum profundus, flexor digitorum superficialis, palmaris longus.

**Lesser-stretched muscles:** Flexor carpi ulnaris, flexor digiti minimi brevis, flexor pollicis longus, extensor carpi radialis brevis, extensor carpi radialis longus, extensor pollicis brevis.

## Commentary

The closer the hands are to the knees, the easier it is to keep the palms on the floor. The farther the hands are in front of the knees, however, the greater the applied stretch. The distance each hand is away from the body's midline also influences stretch intensity. The farther away from the midline, the greater the stretch.

## Wrist Ulnar Deviator and Flexor Stretch



## Technique

Kneel on the floor.

Flex both wrists, and place the palm of each hand on the floor.

Point the fingers medially (the fingertips of the opposing hands pointing toward each other).

While keeping the elbows straight, lean backward (buttocks to the heels), keeping the palms on the floor.

## Muscles Stretched

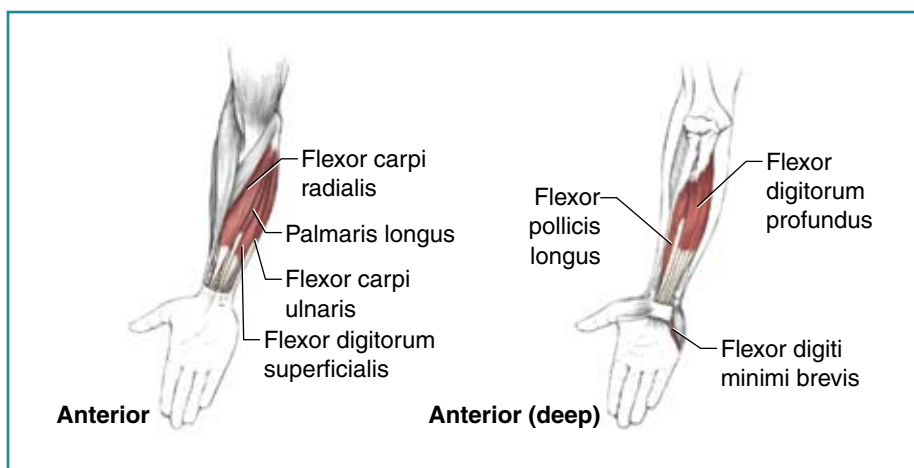
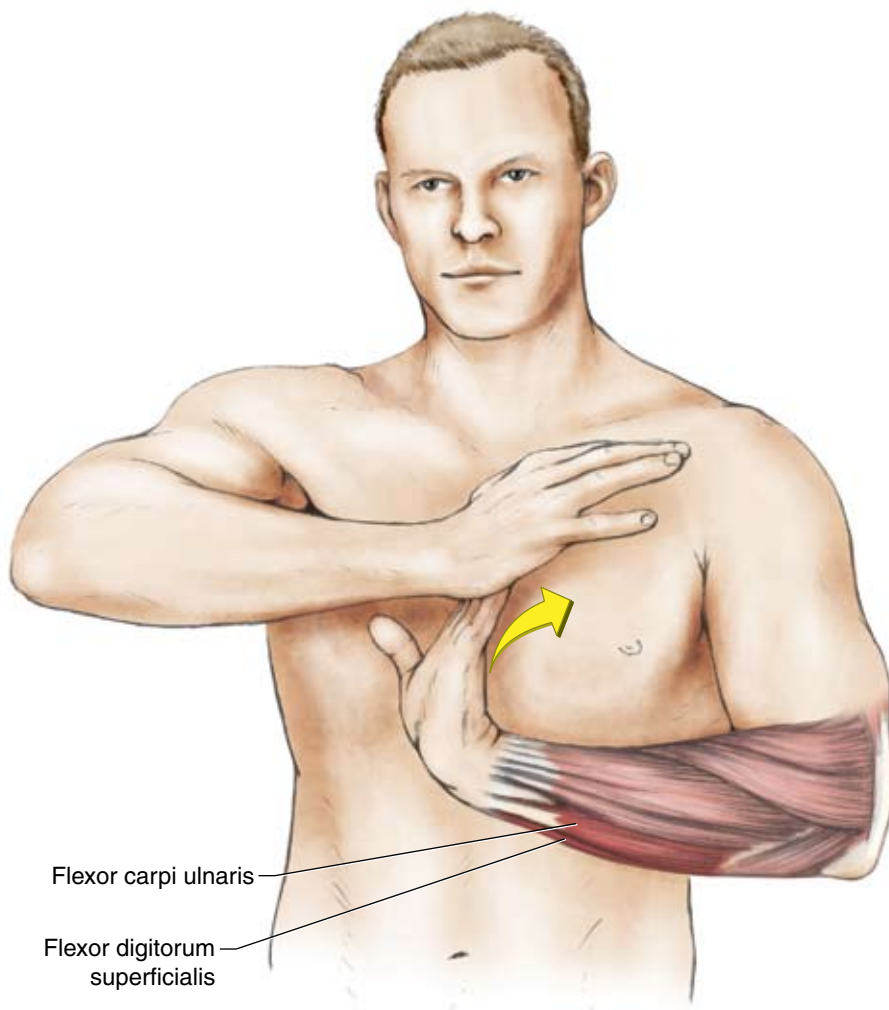
**Most-stretched muscles:** Flexor carpi ulnaris, flexor digitorum profundus, flexor digitorum superficialis, palmaris longus.

**Lesser-stretched muscles:** Flexor carpi radialis, flexor digiti minimi brevis, flexor pollicis longus, extensor carpi ulnaris.

## Commentary

The closer the hands are to the knees, the easier it is to keep the palms on the floor. The farther the hands are in front of the knees, however, the greater the applied stretch. The distance each hand is away from the body's midline also influences stretch intensity. The farther away from the midline, the greater the stretch.

## Finger Flexor Stretch



## Technique

Sit or stand upright.

Flex the elbow at a 90-degree angle, and extend the wrist as far as possible.

Point the fingers upward.

With the right hand, push the fingers on the left hand toward the elbow.

## Muscles Stretched

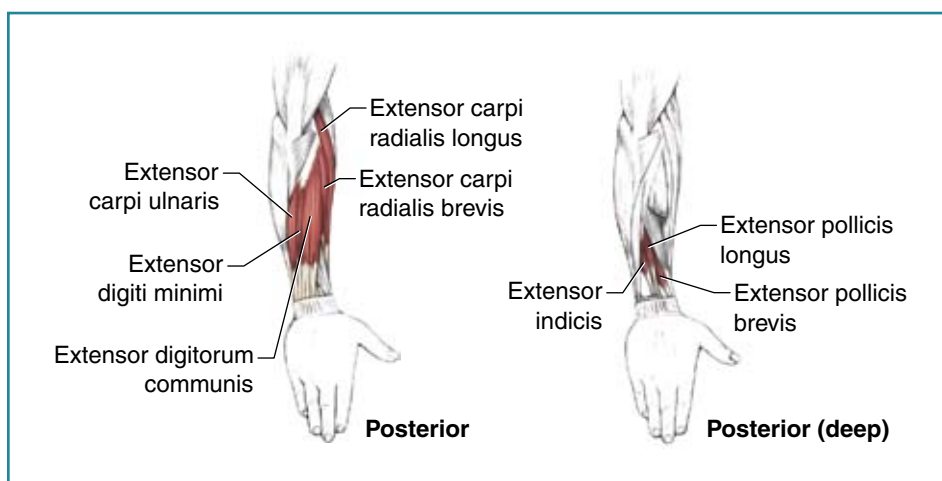
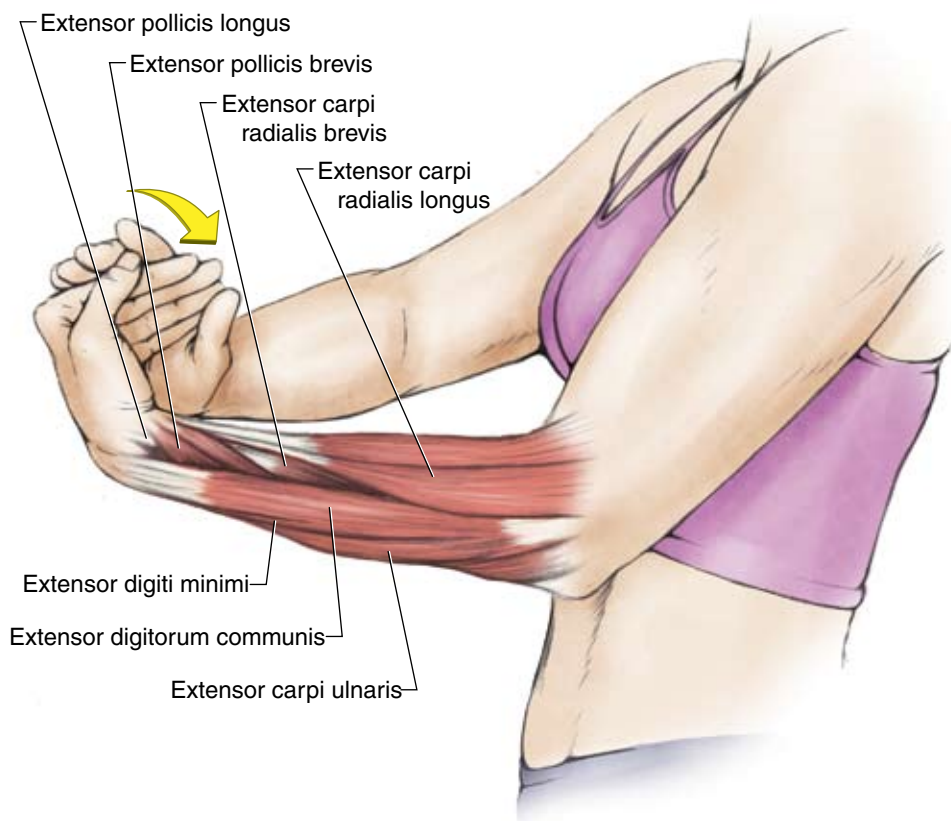
**Most-stretched muscles:** Left flexor carpi radialis, left flexor carpi ulnaris, left flexor digiti minimi brevis, left flexor digitorum profundus, left flexor digitorum superficialis, left palmaris longus.

**Lesser-stretched muscles:** Left flexor pollicis longus.

## Commentary

The elbow angle does not need to be precisely 90 degrees. Choose a comfortable angle. Some people find that fully flexing the elbow makes it easier to push on the hand. With the elbow fully flexed, the push is more downward than across.

## Finger Extensor Stretch



## Technique

Sit or stand upright.

Turn the left arm so that the palm faces up and flex the elbow to a 90-degree angle.

Flex the wrist to a 90-degree angle, and flex the fingers so that they are pointed toward the elbow.

Place the right hand on top of the fingers and press the fingers down toward the forearm.

## Muscles Stretched

**Most-stretched muscles:** Left extensor carpi radialis brevis, left extensor carpi radialis longus, left extensor carpi ulnaris, left extensor digitorum communis, left extensor digiti minimi, left extensor indicis.

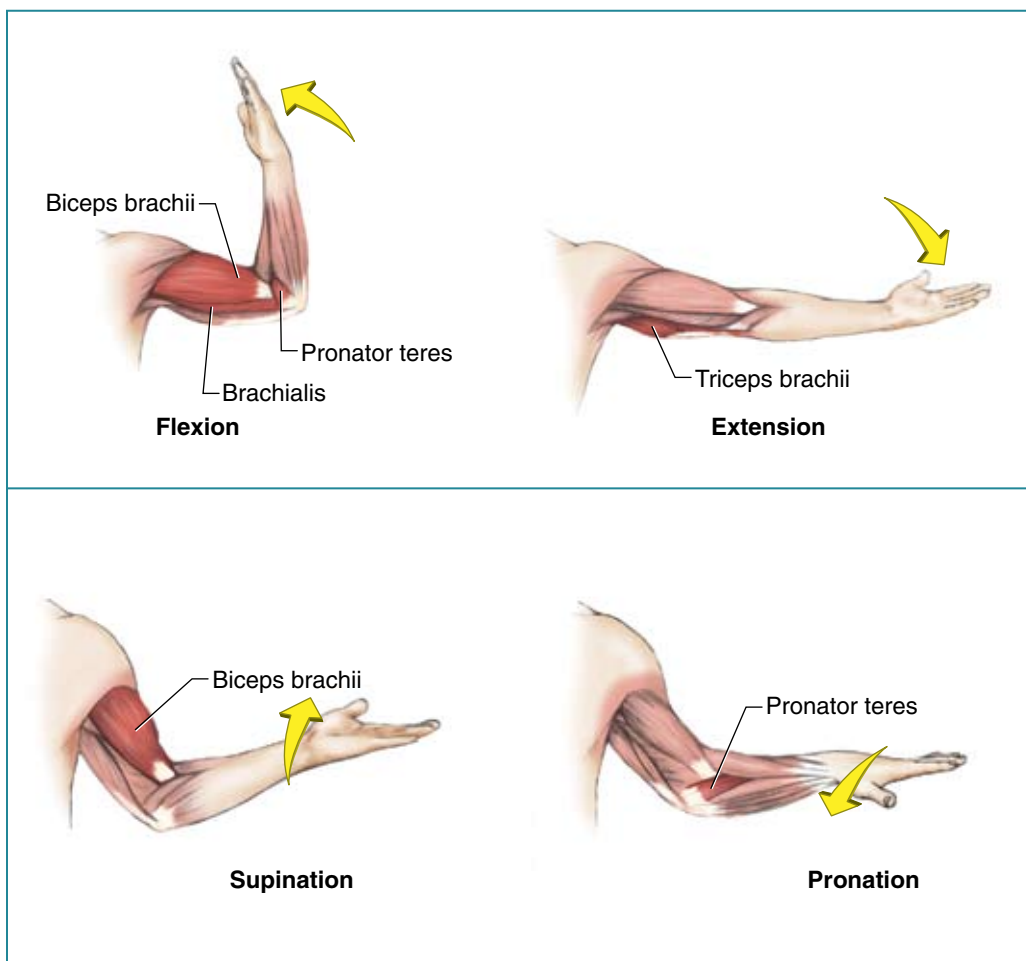
**Lesser-stretched muscles:** Left extensor pollicis brevis, left extensor pollicis longus.

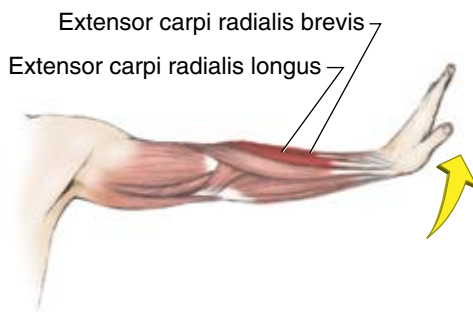
## Commentary

Increase the magnitude of the stretch by flexing the fingers (make a fist). Also, the elbow angle does not need to be precisely 90 degrees. Choose a comfortable angle. Some people find that fully flexing the elbow makes it easier to push on the hand. With the elbow fully flexed, the push is more downward than across.

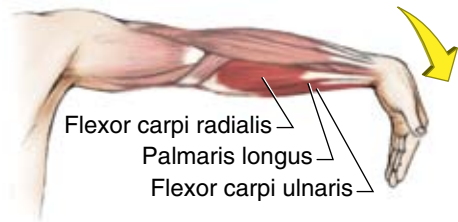
## Arm, Wrist, and Hand Muscle Movements

The stretches in this chapter are excellent overall stretches; however, not all of these stretches may be completely suited to each person's needs. The muscles involved in the various arm and hand movements appear in the table on page 52. To stretch specific muscles, the stretch must involve one or more movements in the opposite direction of the desired muscle's movements. For example, if you want to stretch the flexor carpi radialis, you could perform a movement that involves wrist extension and radial deviation. When a muscle has a high level of stiffness, you should use fewer simultaneous opposite movements (for example, to stretch a very tight flexor carpi radialis, you could start by doing only radial deviation). As a muscle becomes loose, you can incorporate more simultaneous opposite movements.

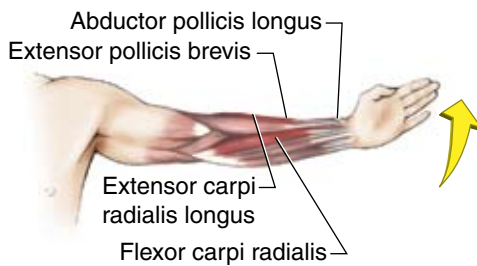




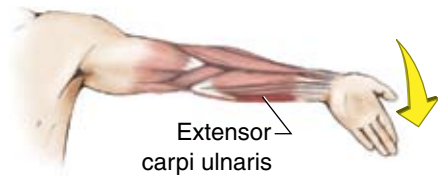
**Wrist extension**



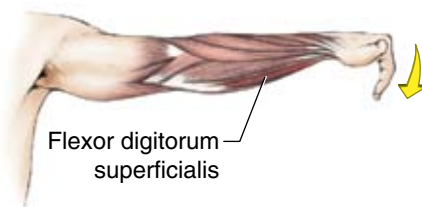
**Wrist flexion**



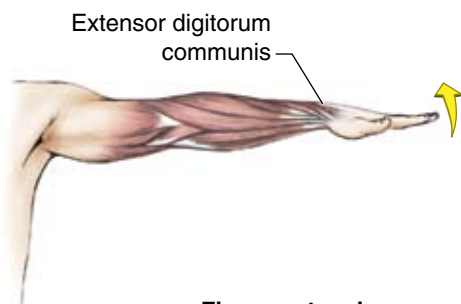
**Radial deviation**



**Ulnar deviation**



**Finger flexion**



**Finger extension**

## ARMS AND WRISTS

| Muscle                         | Elbow flexion | Elbow extension | Pronation | Supination | Wrist flexion | Wrist extension | Radial deviation | Ulnar deviation |
|--------------------------------|---------------|-----------------|-----------|------------|---------------|-----------------|------------------|-----------------|
| Abductor pollicis longus       |               |                 |           |            |               |                 | ✓                |                 |
| Anconeus                       |               | ✓               |           |            |               |                 |                  |                 |
| Biceps brachii                 | ✓             |                 |           | ✓          |               |                 |                  |                 |
| Brachialis                     | ✓             |                 |           |            |               |                 |                  |                 |
| Brachioradialis                | ✓             |                 |           |            |               |                 |                  |                 |
| Extensor carpi radialis brevis |               |                 |           |            |               | ✓               | ✓                |                 |
| Extensor carpi radialis longus |               |                 |           |            |               | ✓               | ✓                |                 |
| Extensor carpi ulnaris         |               |                 |           |            |               | ✓               |                  | ✓               |
| Extensor digitorum communis    |               |                 |           |            |               | ✓               |                  |                 |
| Extensor pollicis brevis       |               |                 |           |            |               |                 | ✓                |                 |
| Flexor carpi radialis          |               |                 |           |            | ✓             |                 | ✓                |                 |
| Flexor carpi ulnaris           |               |                 |           |            | ✓             |                 |                  | ✓               |
| Palmaris longus                |               |                 |           |            | ✓             |                 |                  |                 |
| Pronator quadratus             |               |                 | ✓         |            |               |                 |                  |                 |
| Pronator teres                 | ✓             |                 | ✓         |            |               |                 |                  |                 |
| Supinator                      |               |                 |           | ✓          |               |                 |                  |                 |
| Triceps brachii                |               | ✓               |           |            |               |                 |                  |                 |

## FINGERS

| Muscle                         | Flexion | Extension |
|--------------------------------|---------|-----------|
| Extensor digitorum communis    |         | ✓         |
| Extensor digiti minimi         |         | ✓         |
| Extensor indicis               |         | ✓         |
| Flexor digiti minimi brevis    | ✓       |           |
| Flexor digitorum profundus     | ✓       |           |
| Flexor digitorum superficialis | ✓       |           |
| Flexor pollicis longus         | ✓       |           |